

Study ID	Title	Authors	Year	Study Type	Provenance	I1	I2	E1	E2	E3	Notes
P5	Green AI: Enhancing Sustainability Dash		2025	empirical	search	Y	Y				Backward-snowball SOURCE: 32 refs, 28 excluded at title, 4 full-screened (rows below); Luo = dup of P4. Forward: pending.
S1	The power of training: how efficient GatoStar et al.		2024	empirical	backward snowb	Y		Y			Training-configuration energy (existing concept). Excluded after full-text read.
S4	Towards energy-efficiency by navi/Tian et al.		2024	empirical	backward snowb	N					Inference energy-latency-accuracy trade-off (PI6). EXCLUDED after full-text — I1: XR scene-reconstruction (TSDF) pipeline, not the footprint of an AIML model.
—	A comprehensive review of model Dantas et al.		2024	review	backward snowb	N	Y	Y		Y	Excluded E3 (secondary/review) — Related Work.
—	Deep Compression: Compressing Han et al.		2016	technique	backward snowb	N	N				Excluded I1 (compression technique, does not address environmental impact).
P7	Uncovering Energy-Efficient Practi Yarity et al.		2023	empirical	search	Y	Y				Backward-snowball SOURCE: 42 refs, -9 dedupe (already in corpus); -24 excluded at title, full-screened below. Forward: pending.
S2	Energy and Policy Considerations Strubell; Ganesh; McCallum		2019	empirical	backward snowb	Y	Y	Y			RQ1 energy/carbon/cost; training. Report training cost & HP sensitivity.
S3	On the Dangers of Stochastic Par Bender; Gebre; McMillan-Mayo		2021	position	backward snowb	Y	Y				RQ1 LLM environmental cost; data/training; question scale.
S4	Sustainable AI: Environmental imp Wu et al.		2022	analysis	backward snowb	Y	Y				RQ1 operational-embodied: inference at scale.
—	Designing energy-efficient CNNs i Yang; Chen; Zhe		2017	empirical	backward snowb	Y	Y				RQ3 energy-aware pruning: model design.
S6	Analysing the energy impact of diff Gutierrez; Moraga; Garcia		2022	empirical	backward snowb	Y	Y				RQ3 energy impact of optimisations.
S7	Evaluating the energy efficiency of Li; Chen; Beechi; Zong		2016	empirical	backward snowb	Y	Y				RQ2 hardware energy measurement (CPU/GPU).
—	Mining the ROS ecosystem for gre Malavolta et al.		2021	empirical	backward snowb	N	N				Excluded I1 (robotics software, not AI footprint).
—	Greencaster: training software ene Chowdhury; Borle; Romansky;		2019	empirical	backward snowb	N	N				Excluded I1: software energy models (not AI). Empirical Software Engineering.
—	An exploratory study on assessing Chowdhury; Di Nardo; Hindle ;		2018	empirical	backward snowb	N	N				Excluded I1: Android app energy (not AI).
—	Performance-based guidelines for Cruz; Abreu		2017	empirical	backward snowb	N	N				Excluded I1: mobile-app energy (not AI). MOBILESof.
—	Catalog of energy patterns for mot Cruz; Abreu		2019	empirical	backward snowb	N	N				Excluded I1: mobile-app energy patterns (not AI).
—	Leafactor: Improving energy effice Cruz; Abreu; Rouvignac		2017	tool	backward snowb	N	N				Excluded I1: Android app energy (not AI).
—	On the runtime and energy perform De Macedo; Abreu; Pereira; Sa		2021	empirical	backward snowb	N	N				Excluded I1: WebAssembly/JS runtime energy (not AI).
—	Mining energy-greedy API usage p Linares-Vazquez; Bavota; Bern		2014	empirical	backward snowb	N	N				Excluded I1: Android API energy (not AI).
—	CEMMA: multi-objective optimiz Linares-Vazquez; Bernal-Caride		2017	empirical	backward snowb	N	N				Excluded I1: Android GUI energy (not AI).
P9	Green AI	Schwartz et al.	2020	position	search	Y	Y				Backward-snowball SOURCE: 59 refs. Dedupe (cited by P9 → provenance for S16 Strubell, S31 Henderson, S6 Lacoste) -50 excluded at title (Red-AI example models/datasets/benchmarks + efficient-architecture techniques, I1). 3 full-screened & excluded (below). 0 new pri
S8	Towards the Systematic Reporting Henderson; Hu; Romoff; Bruns		2020	tool	backward snowb	Y	Y				RQ2 experiment-impact-tracker, -30x by region.
S9	Quantifying the carbon emissions i Lacoste; Luccioni; Schmidt; Da		2019	tool	backward snowb	Y	Y				RQ2 ML Emissions Calculator.
—	AI and compute Amodei; Hernandez		2018	report (blog)	backward snowb	N	Y	Y			Excluded E1 (OpenAI blog / grey literature; documents compute-growth trend, not peer-reviewed).
—	Show Your Work: Improved Report Dodge; Gururaman; Card; Scr		2019	empirical	backward snowb	N	N				Excluded I1 (reproducibility / experiment-cost reporting, not environmental footprint).
—	Tackling climate change with modl Reutskiy et al.		2019	position	backward snowb	N	Y				Excluded I1E4 (AI "net" climate — AI for sustainability, not footprint of AI).
P10	Green AI: A Preliminary Empirical i Alzadeh; Castor		2024	empirical	search	Y	Y				Backward-snowball SOURCE: 38 refs. Dedupe (already in sheet): Georgiou (P11), Gutierrez (S25), Henderson (S31), Lacoste (S6), Schwartz (P9), Strubell (S16), Verdecchia-DC (P21), Yarity (P7), Garcia-Martin (S19, excluded E3). Included: 3 (S2, S18, S35). To verify (prepr
S10	Carbontracker: Tracking and Pred Anthony; Kanding; Selvan		2020	tool	backward snowb	Y	Y				RQ2 tool: predict/track kWh & CO2e.
S11	Compute and energy consumption Desislavov; Martinez-Plumed; I		2021	analysis	backward snowb	Y	Y				RQ1/RQ2 INFERENCE energy trends. Resolves pending-S18 (cited by included primary P10). Published in Sustainable Computing.
S12	A Study on the Battery Usage of D Jacques; Alzadeh; Castor		2024	empirical	backward snowb	Y	Y				RQ1 energy: on-device/runtime inference. MOBILESof. Concept-redundant (runtime energy).
—	Energy Efficiency of Training Neu Xu; Martinez-Fernandez; Marti		2023	empirical	backward snowb	?	Y				MAYBE: arXiv 2022.09567. RQ3 training/infrastructure energy. Include only if a peer-reviewed version exists (else E1).
—	Energy Efficiency Considerations i Fischer; Jakob; Mark		2023	empirical	backward snowb	?	Y				Lean EXCLUDE E1 (arXiv preprint, concept-redundant benchmark energy). Verify venue.
—	FECOM: A Step Towards Fine-Grai Rajput; Widmayer; Shang; Kec		2023	tool	backward snowb	?	Y				Lean EXCLUDE E1 (arXiv preprint; RQ2 measurement, concept-redundant). Verify venue.
P11	Green AI: Do Deep Learning Fram Georgiou; Kechagia; Sharma; i		2022	empirical	search	Y	Y				Backward-snowball SOURCE: -85 refs (56 green-software-heavy). Dedupe: Schwartz (P9), Strubell (S16), Li (S34), Garcia-Martin (S19, excl E3). Included: 2 (S36, S37). Rest excluded: large green-software cluster (Cruz/Hindle/Pinto/Castor/Pereira/Mantos — app/software e
S13	An Analysis of Deep Neural Netw Canziani; Paszke; Culurciello		2016	analysis	backward snowb	Y	Y				RQ1/RQ2 power, accuracy & memory of DNN models (trade-off, P16). arXiv, foundational (-well-cited).
—	Energy-Efficient Machine Learning Kumar; Zhang; Lu; Wang; Shi		2020	empirical	backward snowb	N	Y				contribution (JEPD) is a general Java CODE energy profiler = green "software" engineering (I1 concern, consistent with our GreenScaler/Java energy exclusions); also part survey-summary (E3 tension). Lean: weak-include as edge measurement practice, or EXCLUDE if stict
—	GreenBundler: An Empirical Study i Chowdhury; Hindle; Kazman; S		2019	empirical	backward snowb	N	N				Excluded I1: web/software bundled-processing energy (not AI).
—	Green Software Engineering Done Cruz		2021	blog	backward snowb	N	Y				Excluded E1 (blog) + I1 (green software).
—	Tools to Measure Software Energy Cruz		2021	blog	backward snowb	N	Y				Excluded E1 (blog) + I1 (green software).
—	Estimating Software Energy Conx; Fu; Qian; Luan		2018	empirical	backward snowb	N	N				Excluded I1: ML used to estimate software energy (not footprint of AI).
—	An Empirical Study of Practitioners Manotas; Bird; Zhang; Shephe		2016	empirical	backward snowb	N	N				Excluded I1: green software engineering (not AI).
—	Mining Energy-Aware Commits Moura; Pinto; Ebert; Castor		2015	empirical	backward snowb	N	N				Excluded I1: software-commit energy (not AI).
—	What Do Programmers Know about Pang; Hindle; Adams; Hussain		2016	empirical	backward snowb	N	N				Excluded I1: software energy (not AI).
—	Fine-Grained Power Mining for i Pathak; Hu; Zhang; Bark; Wang		2011	empirical	backward snowb	N	N				Excluded I1: smartphone power modeling (not AI).
—	The Influence of the Java Collectio Pereira; Couto; Saravia; Cunha		2016	empirical	backward snowb	N	N				Excluded I1: small/software energy (not AI).
—	RAPL: Memory Power Estimation i David; Gorbatov; Hanebutte; K		2010	method	backward snowb	N	N				Excluded I1: general hardware power measurement (not AI).
—	A Validation of DRAM RAPL Powe Desrochers; Paradis; Weaver		2016	method	backward snowb	N	N				Excluded I1: general power measurement (not AI).
—	Rapid and accurate energy model Kavanagh; Djemame		2019	method	backward snowb	N	N				Excluded I1: general energy models (not AI).
—	RAPL in Action: Experiences in Us Khan; Hrkic; Niemi; Numminen; i		2016	method	backward snowb	N	N				Excluded I1: general power measurement (not AI).
—	Dynamic Voltage Frequency Scal Coles-Ruiz; Prieto; Garcia-Gal		2017	tool	backward snowb	N	N				Excluded I1: general green-cloud/DVS (not AI).
P15	Energy-Efficient ML Architecture: i Sharma; Mathur		2025	framework	search	Y	Y				Backward-snowball SOURCE: 14 refs. Dedupe: Schwartz (P9), Verdecchia review (P1, RW), Barberato (P3, RW). Included: 4 (S38-S41 — all concept-redundant, pending saturation rule). Excluded title-passers: 7. Forward: pending.
—	Lightweight and Energy-Efficient D Kim; Kim; Yoo		2023	empirical	backward snowb	N	Y				this is an efficiency-TECHNIQUE / hardware-accelerator DESIGN paper (build a low-energy accelerator), not a study OF AI's environmental impact → consistent with our I1 exclusion of MobileNets/quantization/accelerator technique papers. Lean: EXCLUDE (I1) under strict re
—	Optimizing Transformers for Energ Paula et al.		2025	empirical	backward snowb	Y	Y		Y		RQ3 transformer energy optimization (inference). Scientific Reports. Concept existing. EXCLUDED — E2: full text not obtainable; concept-redundant (efficient transformers). Reinstated if PDF found.
S14	Energy Consumption Index for Dei Aquino-Britzel et al.		2025	empirical	backward snowb	Y	Y				RQ2 energy metric/index. IEEE Access. Concept existing.
S15	A Holistic Framework for Green AI Tabataba et al.		2024	framework	backward snowb	Y	Y				RQ3 green-AI Framework. Sustainable Computing. Concept existing.
—	DistillBERT: A Distilled Version of B Santhi et al.		2019	technique	backward snowb	N	N				Excluded I1 (distillation technique).
—	TinyBERT: Distilling BERT for Natuliao et al.		2020	technique	backward snowb	N	N				Excluded I1 (distillation technique).
—	Deep Compression Han; Mao; Dally		2016	technique	backward snowb	N	N				Excluded I1 (compression technique).
—	A Survey of Quantization Techniq Gholami et al.		2021	review	backward snowb	N	N		Y		Excluded E3 (survey) + I1 (technique).
—	Improving Automated Machine Lei Castellanos-Nieves; Garcia-Fo		2023	empirical	backward snowb	N	Y	Y			On-topic but concept-redundant (AutoML metrics). = P39 (already screened). Not added.
—	Data Pruning and Active Learning i Scala et al.		2024	empirical	backward snowb	N	Y	Y			On-topic but concept-redundant (data-centric). = P38 (already screened). Not added.
—	Energy-Efficient BERT Optimiz iyer; Kumar; Mehta		2025	empirical	backward snowb	N	N	Y			Excluded E1 (weak venue, JRA).
P21	Data-centric Green AI: an explorat Verdecchia; L Cruz; J Salou;		2022	empirical	search	Y	Y				Excludes/dedupes: Strubell (S16), Lacoste (S6), Bender (S20), Schwartz (P9), Wu (S4), Georgiou (P11), Garcia-Martin-estimation (S19), Zhu IoT (=P37, redundant), Amodei (E1), Kohn (E4, ML-for-HPC-energy). Zogaj (I1), large green-software cluster (Cruz+3, Pereira, Oliveira
S16	Sustainable AI: AI for sustainability Van Wynneberghe		2021	conceptual	backward snowb	Y	Y				RQ1 of AI vs for-AI (canonical scope anchor).
S17	Estimation of energy consumption Garcia-Martin; Lavesson; Grah		2019	empirical	backward snowb	Y	Y				RQ1/RQ2 energy hotspots of an ML algorithm. Reconciled from 2017 "energy hotspots" to the 2019 JPDC paper actually held.
S18	Carbon-aware computing for data Radovanovic et al.		2022	empirical	backward snowb	Y	Y				RQ3 carbon-aware scheduling (operational, PM4). IEEE Trans. Power Systems.
S19	Aligning artificial intelligence with c Kaack; Dotti; Strubell; Kamiya;		2022	analysis	backward snowb	Y	Y				RQ1 footprint of AI (+ governance). Nature Climate Change.
P23	Green AI Practices in a Future Per H Kawan; A Kumar; P Rajen		2025	Search	backward snowb	Y	Y				Excludes model-compression techniques/surveys (AMC, EIE, Deep Compression, Distilling, BranchyNet, Mixed Precision, "Survey of Model Compression" +2), models (ResNet/VGG/BERT/GPT-2/ViT/EfficientNet), off-topic (drug nanocarriers, blockchain, etc.).
S20	Carbon Emissions and Large Neu Patterson et al.		2021	analysis	backward snowb	Y	Y				RQ3 AIAs; datacentre/scheduling.
S21	The Carbon Footprint of ML Traini Patterson et al.		2022	analysis	backward snowb	Y	Y				RQ3 4 best practices.
S22	Measuring the Carbon Intensity of Dodge et al.		2022	empirical	backward snowb	Y	Y				RQ2/RQ3 carbon-aware region/time/pause.
S23	Estimating the Carbon Footprint of Luccioni; Viguer; Ligozat		2023	empirical	backward snowb	Y	Y				Full-lifecycle LLM footprint. (cited by included primary P23). JMLR.
S24	Energy and Carbon Consideration Wang; Na; Strubell; Frieder; Li		2023	empirical	backward snowb	Y	Y				RQ1/RQ3 fine-tuning energy/carbon (thesis-relevant). EMNLP Findings.
S25	In TinyML, Sustainable? Assessing Prakash; Banerjee; Banerjee; Ma		2023	empirical	backward snowb	Y	Y				RQ1/RQ3 TinyML; embodied and inference footprint (NEW-ish concept).
—	Green Federated Learning Youssefyou et al. (Meta)		2023	empirical	backward snowb	P23	Y	Y			RQ3 federated-learning energy/carbon (NEW-ish concept). Verify venue (arXiv). EXCLUDED after full-text — E1: arXiv-only preprint, not foundational.
—	Mitigating Carbon Footprint for Kvi (authors n.d.)		2023	empirical	backward snowb	Y	Y		Y		RQ3 carbon of KD-compression. PMC journal. EXCLUDED - E2: full text not obtainable; concept-redundant (KD carbon). Reinstated if PDF found.
P25	Green AI-Driven Concept... Detect Acmal et al.		2024	empirical	search	Y	Y				Backward SOURCE: 46 refs. 0 NEW — all pruning/compression techniques (I1), CNN models, or Emlera/biomedical domain; does not cite the green-AI canon. Forward: pending.
P26	Green AI: Optimizing Energy Effici Krishnam		2025	framework	search	Y	Y				Backward SOURCE: 19 refs. 0 NEW primaries. Ashouri "Energy-Efficient DL: A Survey" → Related Work (E3). [9] Resource Mgmt for DL, [10] DL job scheduling excluded I1 (compute scheduling, not environmental). Rest = industry/news reports (E1) + general datacentre ener
P29	Green AI: Balancing Model Compl D Aggarwal; D Sharma; AB Sa		2025	search	backward snowb	Y	Y				
S27	Establishing green AI certifications Gupta; Shankar; Savana		2024	framework	backward snowb	Y	Y		Y		RQ3 reporting/governance (certification framework). IEEE. EXCLUDED — E2: full text not obtainable; concept-redundant (governance framework). Reinstated if PDF found.
S28	Towards Green AI: Current Status Chemi et al.		2024	analysis	search	Y	Y				RQ1/RQ3 LLM; embodied and inference footprint (NEW-ish concept).
—	A Synthesis of Green Architectural Javerpava; Lago; Bogner; Lew		2024	secondary	backward snowb	Y	Y			Y	RQ3 catalog of green architectural tactics for ML systems. ICSE-SEIP FLAG: it's a "synthesis" (possibly E3-secondary → Related Work) but provides an extractable tactics catalog. NB: co-authored by J. Bogner (your supervisor). Secondary study (E3) → Related Work (not a
P30	Position Paper: The Environmental Fousti		2025	position	search	Y	Y				Backward SOURCE: -43 refs. 0 NEW - canon dedupes (Strubell S16, Lacoste S6) + efficiency techniques (compression/quantization/distillation/LsRA/adapters/pruning/NAS, I1) + surveys (edge/datacentre/approximate computing, E3) + models (GPT-3, ViT, T5). Forward: peni